

1. **Entities and attributes:** Entity → table, identifier → primary key · Composite and derived attributes · Multivalued attribute → its own table
2. **Binary relationships:** 1:N relationship → foreign key on the N side · M:N relationship → its own table · 1:1 relationship → the key passes to one side · Participation → NOT NULL on the foreign key
3. **Special cases:** Weak entity → owner's key + partial key · Unary (recursive) relationship · Ternary relationship → a table with three FKs
4. **Hierarchies:** Hierarchy: the three strategies · Choosing a hierarchy strategy

Entities and attributes

Entity → table, identifier → primary key

Every entity type becomes a **table** with the same name, and its identifier becomes the table's **primary key**.

- Rule 1: **entity** → **table**, named after the entity.
- Rule 2: **attribute** → **column** of that table.
- Rule 3: **identifier** → **primary key**.

Resulting tables

Book (ISBN, title, year)

Underlined: primary key. Italic with →: foreign key. ? marks a foreign key that can be NULL.

Book
<u>ISBN</u>
title
year

Example. The entity Book with ISBN, title and year becomes Book(ISBN, title, year). One row per book, and the ISBN identifies it.

Common mistake. Adding an artificial id column when the entity already has an identifier. If the model says the ISBN identifies a book, the ISBN is the primary key.

Composite and derived attributes

A **composite** attribute is split into one column per part. A **derived** attribute is left out, because it can be computed.

- **Composite** → its parts become columns; the composite name itself disappears.
- **Derived** → no column; compute it in queries.

Resulting tables

Customer (customer id, name, birth date, street, city, postal code)

Underlined: primary key. Italic with →: foreign key. ? marks a foreign key that can be NULL.

Customer
<u>customer id</u>
name
birth date
address (street, city, postal code)
/age

Example. Customer(customer id, name, birth date, address(street, city, postal code), /age) becomes Customer(customer id, name, birth date, street, city, postal code).

Common mistake. Keeping an age column next to birth date. After a year the stored age is wrong, while the birth date is still right.

Multivalued attribute → its own table

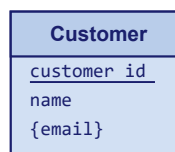
A **multivalued** attribute becomes a new table. Its primary key is the owner's key plus the value.

- New table, usually named owner + attribute (CustomerEmail).

- Columns: the owner's key (**PK and FK**) + the value (**PK**).
- The owner's table keeps its other attributes and loses the multivalued one.

Resulting tables

Customer (customer id, name)
CustomerEmail (customer id→Customer, email)



Underlined: primary key. Italic with →: foreign key. ? marks a foreign key that can be NULL.

Example. Customer(customer id, name, {email}) becomes Customer(customer id, name) and CustomerEmail(customer id → Customer, email).

Common mistake. Adding the columns email1, email2, email3. That sets an arbitrary limit, leaves NULLs and makes searching for an e-mail look in three places.

Binary relationships

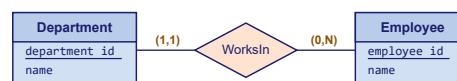
1:N relationship → foreign key on the N side

A one-to-many relationship gets **no table of its own**. The key of the "one" side travels to the "many" side as a **foreign key**.

- Rule 5: 1:N → **FK on the N side**, referencing the 1 side.
- The FK is **NOT NULL** when the minimum at the 1 side is 1 (see "Participation").

Resulting tables

Department (department id, name)
Employee (employee id, name, *department id*→Department)



Underlined: primary key. Italic with →: foreign key. ? marks a foreign key that can be NULL.

Example. Department (1,1) — WorksIn — (0,N) Employee becomes Employee(employee id, name, department id → Department).

Common mistake. Putting the foreign key on the "one" side, such as an employee id in Department. It can only hold one employee per department.

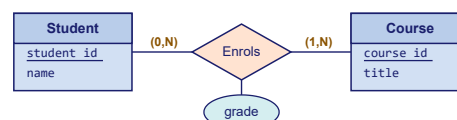
M:N relationship → its own table

A many-to-many relationship becomes a **new table**. Its primary key combines the keys of both entities, and each of them is also a foreign key.

- Rule 4: M:N → **table** named after the relationship.
- PK = **both foreign keys together**.
- The relationship's attributes become columns of this table.

Resulting tables

Student (student id, name)
Course (course id, title)
Enrols (student id→Student, course id→Course, grade)



Underlined: primary key. Italic with →: foreign key. ? marks a foreign key that can be NULL.

Example. Student (0,N) — Enrols(grade) — (1,N) Course becomes Enrols(student id → Student, course id → Course, grade).

Common mistake. Making only one of the two foreign keys the primary key. Then a student could appear in a single course only.

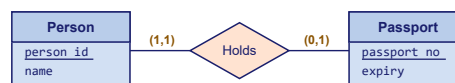
1:1 relationship → the key passes to one side

In a one-to-one relationship the key of **either** side can pass to the other. With (0,1)/(1,1), the course rule passes it to the **optional** side.

- Rule 6: 1:1 → the key of either side passes to the other.
- Rule 7: with (0,1)/(1,1), the key passes to the optional side (the one with min 0 at the opposite end).
- Use the switch below to see both accepted designs.

Resulting tables

Person (person id, name, *passport no*→Passport?)
Passport (passport no, expiry)



Underlined: primary key. Italic with →: foreign key. ? marks a foreign key that can be NULL.

Example. Person (1,1) — Holds — (0,1) Passport becomes Person(person id, name, passport no → Passport), with the FK allowed to be NULL.

Common mistake. Merging both entities into one table. It is sometimes acceptable, but it is not the course rule, and it leaves empty columns for every person without a passport.

Participation → NOT NULL on the foreign key

The **minimum** cardinality decides whether a foreign key can be NULL. Read it **across** the relationship, at the end the FK points to.

- Min 1 at the referenced side → FK **NOT NULL**.
- Min 0 at the referenced side → FK **can be NULL** (shown with ? in the notation).
- Primary-key columns are always NOT NULL.

Resulting tables

Customer (customer id, name)
Order (order id, date, *customer id*→Customer, *courier id*→Courier?)
Courier (courier id, name)



Underlined: primary key. Italic with →: foreign key. ? marks a foreign key that can be NULL.

Example. Customer (1,1) — Places — (0,N) Order and Courier (0,1) — Delivers — (0,N) Order become Order(order id, date, customer id → Customer NOT NULL, courier id → Courier NULL).

Common mistake. Reading the cardinality on the wrong side. The card next to Order says how many orders a customer has. It decides nothing about the FK in Order.

Special cases

Weak entity → owner's key + partial key

A **weak entity** cannot be identified on its own. Its primary key is the **owner's key** (also a foreign key) plus its **partial key**.

- PK = owner key (FK) + partial key.
- The FK is always NOT NULL, because it is part of the PK.
- Deleting the owner usually deletes its weak rows (ON DELETE CASCADE).

Resulting tables

Building (building id, address)

Room (*building id*→Building, number, floor)



Underlined: primary key. Italic with →: foreign key. ? marks a foreign key that can be NULL.

Example. Building (1,1) = Contains = (1,N) Room(number, floor) becomes Room(building id → Building, number, floor).

Common mistake. Using the partial key alone as the primary key. Two buildings with a room 101 would then collide.

Unary (recursive) relationship

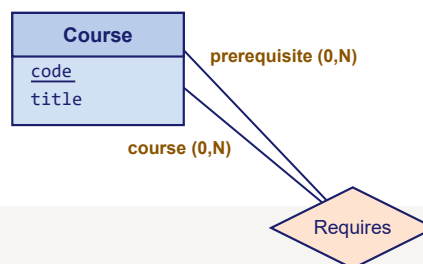
When an entity relates to **itself**, apply the usual 1:N or M:N rule. The foreign keys point to the **same table** and are named after the **roles**.

- Unary 1:N → a FK to the **same table**, named after the role.
- Unary M:N → a **table with two FKs** to the entity, one per role.

Resulting tables

Course (code, title)

Requires (*prerequisite*→Course, *course*→Course)



Underlined: primary key. Italic with →: foreign key. ? marks a foreign key that can be NULL.

Example. Course (0,N) as prerequisite — Requires — (0,N) as course becomes Requires(prerequisite → Course, course → Course).

Common mistake. Creating a second table for the same entity, such as Mentor next to Person. A mentor is a person, so the role is a column, not a table.

Ternary relationship → a table with three FKs

A relationship among **three** entities always becomes its own table, with a foreign key to each of them.

- Ternary → **table with one FK per entity**.
- PK: the FKs of the "many" ends (all three if every end is N).

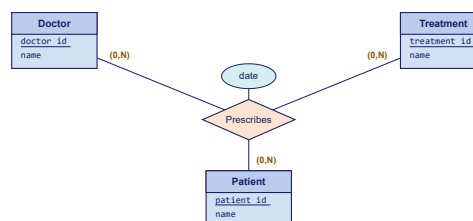
Resulting tables

Doctor (doctor id, name)

Treatment (treatment id, name)

Patient (patient id, name)

Prescribes (*doctor id*→Doctor, *treatment id*→Treatment, *patient id*→Patient, date)



Underlined: primary key. Italic with →: foreign key. ? marks a foreign key that can be NULL.

Example. Doctor, Patient, Treatment — Prescribes(date) becomes Prescribes(doctor id → Doctor, patient id → Patient, treatment id → Treatment, date).

Common mistake. Splitting the ternary into three binary relationships. You lose which doctor prescribed which treatment to which patient.

Hierarchies

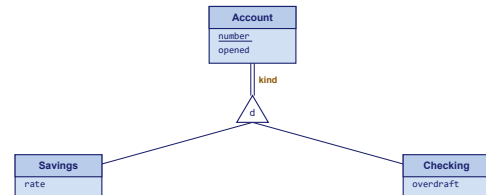
Hierarchy: the three strategies

A generalization (supertype and subtypes) can be turned into tables in **three** ways. Use the switch below to see the tables each one produces.

- Supertype + subtypes: no NULLs; reading a full row needs a join.
- Single table: no joins; NULLs in the subtype columns.
- Subtypes only: no joins and no NULLs, but the supertype cannot be referenced as a whole.

Resulting tables

Account (number, opened)
Savings (number→Account, rate)
Checking (number→Account, overdraft)



Underlined: primary key. Italic with →: foreign key. ? marks a foreign key that can be NULL.

Example. Account(number, opened) is either a Savings(rate) or a Checking(overdraft) account, so the hierarchy is total and disjoint.

Common mistake. Choosing "subtypes only" for a partial hierarchy. Accounts that are neither savings nor checking would have nowhere to go.

Choosing a hierarchy strategy

All three strategies are correct. Pick one by how the data is **used**: shared relationships, how many subtype attributes there are, and whether the hierarchy is total.

Trade-offs of the three strategies

Strategy	NULLs	Joins to read a row	Needs a total hierarchy
Supertype + subtypes	No	Yes (supertype ⋈ subtype)	No
Single table	Yes, in the subtype columns	No	No
Subtypes only	No	No (but a UNION to list them all)	Yes

Common mistake. Leaving out the discriminator in the single-table strategy. Without it a row with every subtype column NULL cannot be classified.